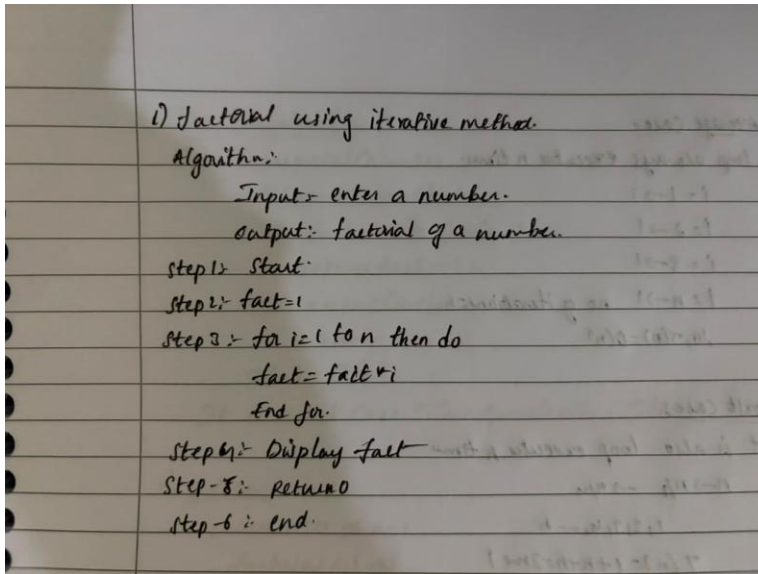


PRACTICAL-04

AIM: Implementation and Time analysis of factorial program using iterative and recursive method

1.) ITERATIVE METHOD

ALGORITHM:



PROGRAM:

```
#include <iostream>

using namespace std;

int main(){

    int n;

    int fact = 1;

    cout << "enter a number:";

    cin >> n;

    for(int i = 1; i <= n; i++){

        fact = fact * i;

    }

    cout << "Factorial of " << n << " = " << fact;

    return 0;

}
```

OUTPUT:

```
Output
enter a number:5
Factorial of 5 = 120

=== Code Execution Successful ===
```

TIME COMPLEXITY:

Time Complexity:
fact = 1
for i = 1 to n
fact = fact * i
 $T(n) = 1 + n + n$
 $= 2n + 1$
 $T(n) = O(n)$

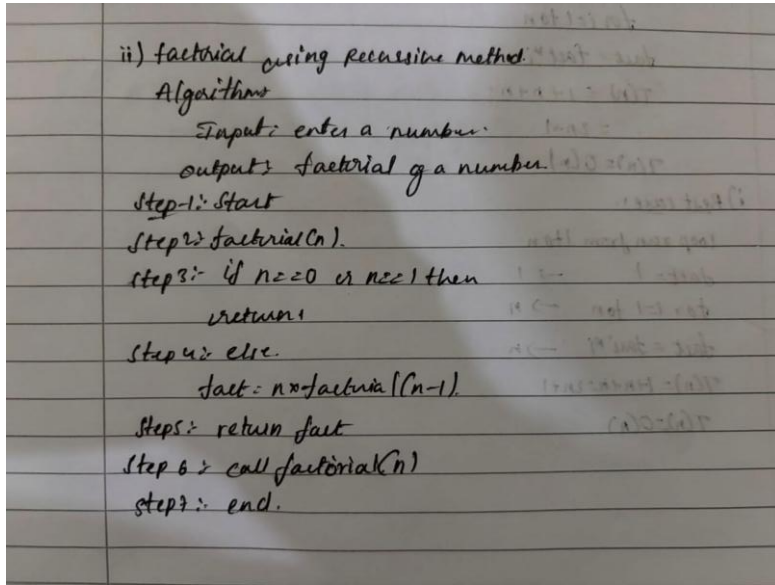
i) Best case:
loop run from 1 to n
fact = 1 $\rightarrow 1$
for i = 1 to n $\rightarrow n$
fact = fact * i $\rightarrow n$
 $T(n) = 1 + n + n = 2n + 1$
 $T(n) = O(n)$

Average case:
loop always executes n times.
i = 1 $\rightarrow 1$
i = 2 $\rightarrow 1$
i = 3 $\rightarrow 1$
i = n $\rightarrow 1$ no of iterations = n
So, $T(n) = O(n)$

Worst case:
it is also loop executes n times
 $n \rightarrow n/2 \rightarrow n/4$
1, 2, 3, 4, ... n
 $T(n) = 1 + n + n = 2n + 1$
 $T(n) = O(n)$
Best = Average = Worst = $O(n)$

2.) RECURSIVE METHOD:

ALGORITHM:



PROGRAM:

```
#include <iostream>

using namespace std;

class factorial{

public:

    int factr(int n){

        if(n == 0 || n == 1){

            return 1;

        }

        else{

            int result = n*factr(n-1);

            return result;

        }

    }

};

int main(){

    int n,result;

    factorial obj;
```

```

cout << "enter a number:";

cin>>n;

result = obj.factr(n);

cout << "factorial of " << n << " is " << result;

return 0;

}

```

OUTPUT:

```

Output
enter a number:7
factorial of 7 is 5040

=== Code Execution Successful ===

```

TIME COMPLEXITY:

Time complexity:

factorial(n) $\rightarrow n$
 if (n==0 || n==1) $\rightarrow n$
 $n * \text{factorial}(n-1) \rightarrow (n-1)$
 return fact $\rightarrow n$
 $T(n) = n + (n-1) + n$
 $= 4n-1$
 $T(n) = O(n)$
 It is worst case Time Complexity $= O(n)$

i) Best case:
 when $n=0$ or $n=1$
 factorial(1) $\rightarrow 1$
 $T(n) = O(1)$

ii) Average case
 no of calls is 7
 ex: $n=5 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$
 $T(n) = O(n)$

iii) worst case:
 large number of n continues until it reaches 1
 ex: $n \rightarrow n-1 \rightarrow n-2 \rightarrow \dots \rightarrow 2 \rightarrow 1$
 $T(n) = O(n)$

CONCLUSION: Both iterative and recursive methods are used to find factorial. Both take $O(n)$ time, but the iterative method uses less memory and is simpler to implement.